

Theorem S1 / Proposition S2

$$|C_t^{(N)} - C_t| \leq C \varepsilon_N^2 = C 4^{-N}, \quad C \sim C_0(M, k, L, K) m^2 \Lambda_k(T), \quad \Lambda_0(T) = T$$

(I) vacuum + field (static)

$$|C_t^{(N)} - \tilde{C}_t^{(N)}|$$

Lemma S3

Fock-space lift,
slot factor $(p+2)$

Lemma S4

lattice vacuum
 $\Omega_N \rightarrow \Omega$:
 $\eta_M(N)$ explicit
[v5, Prop. 6.3]

Lemma S1

diagonal L^2 rate
 $\|(\hat{\ell}_k^{(N)} - \ell_k)\xi\| \lesssim \varepsilon_N^2$
[v5, Lem. 4.15]

Lemma S2

off-diagonal
HS on strip
 $0 < |l| < |k|$
(= 0 for $k=0, \pm 1$)

(II) generator + dynamics on $[0, T]$

$$|\tilde{C}_t^{(N)} - C_t|$$

Quantitative Duhamel

[v5, Lem. 4.19]

$$\|(U_t^{(N)} - U_t)\Psi\| \leq \varkappa_N \Lambda_k(t) \|\mathcal{N}^p \Psi\|, \quad \varkappa_N \lesssim \varepsilon_N^2$$

Lemma S5 [v5,

Lem. 4.17, Prop. 6.6]
 N -uniform energy bounds;
energy growth $\|\mathcal{N}^p e^{itH_k} \Psi\| \leq e^{pc_k t} \|\mathcal{N}^p \Psi\|, \quad c_0 = 0$

m -point telescoping: m telescoped terms $\times$
 $O(m)$ energy-shell slot growth $\Rightarrow O(m^2)$